

Math 141, S10
Quiz 1

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1.(6 points) Find the domain of the following functions.

(a) $f(x) = \sqrt{x} + \sqrt{4-x}$

Solution: $x \geq 0$ and $4-x \geq 0$

$$\Rightarrow 0 \leq x \leq 4$$

Thus the domain is $\{x | 0 \leq x \leq 4\}$
(or $[0, 4]$)

(b) $f(x) = \frac{3}{x^2 + x}$

Solution: $x^2 + x \neq 0 \Rightarrow x(x+1) \neq 0 \Rightarrow x \neq 0$ and $x \neq -1$

Thus the domain is $\{x | x \neq 0, -1\}$
(or $(-\infty, -1) \cup (-1, 0) \cup (0, +\infty)$)

Problem 2.(4 points) Sketch the graph of the function $y = 2^x$ in the following:

